Plagiarism Detector .net

Plagiarism Report

0%
Plagiarism

Unique	100%
Exact Match	0%
Partial Match	0%

Primary Sources

Excluded URL (s)

01 None

Content

Data Structures And Analysis

Software Requirements Specification
(SRS) Document

Project Name: Cybershooter

Date: 27/09/2025

Version: Progress Report 1.1

Report By: Samra Mehmood

Course: BEE 61 C

Submitted to: Lt Col Muhammad Imran Javaid

1
Revisions

Version Primary
Author(s)
Description of Version Date
Completed
1.1 Samra Mehmoood
Abdullah Khan

Initial Version

27/09/2025

2

Table of Contents

1. Introduction

1.1 Purpose

The purpose of CyberShooter is to provide an interactive shooting arcade experience where the
player controls a jet, shoots down enemy ships and earns points. The project aims to demonstrate
practical implementation of object-oriented programming and game design principles using
Python and Pygame.

1.2 Document Conventions

This document follows IEEE SRS formatting conventions. Headings are numbered, functional
requirements are listed using numbered subsections, and technical terms are italicized where
needed.

1.3 Intended Audience

This SRS is intended for:
- Testers: To verify the system against documented requirements.
- Project Supervisors/Professors: To review progress and compliance with standards.

1.4 Scope

CyberShooter is a 2D arcade-style shooting game developed in Python using Pygame. The scope
of this software includes designing a player-controlled jet, bullet shooting mechanics, enemy
spawning, collision detection, scoring and game over states. The game provides entertainment,
demonstrates programming skills and serves as a foundation for future game development
projects.

1.5 References

- IEEE Software Requirements Specification Standards
- Python official documentation (https://docs.python.org/3/)
- Pygame documentation (https://www.pygame.org/docs/)

2. General Description

2.1 Product Perspective

CyberShooter is a stand-alone arcade game. It was inspired by traditional space shooter games
but implemented using Python data structures. It does not rely on external software beyond
Pygame and runs locally on a user’s computer.

2.2 Product Features

- Player controls a jet with keyboard inputs.
- Player can shoot bullets to destroy enemies.
- Enemies spawn randomly from the top of the screen.
- Collision detection removes enemies and awards points.
- High score tracking system.
- Game over screen with restart option.

2.3 User Class Characteristics

Users are casual players and students. They require basic computer literacy to use keyboard
controls. No prior gaming experience is necessary.

2.4 Operating Environment
The game runs on Windows, Linux, or macOS with Python 3.x and Pygame installed.
Recommended hardware: 4GB RAM, dual-core processor standard keyboard.
2.5 Constraints
- Limited graphics since Pygame is 2D.
- Performance depends on user's system speed.
- No multiplayer support in current version.
2.6 Assumptions and Dependencies
Assumptions:
- Users have Python and Pygame installed.
- Players understand basic keyboard controls.
Dependencies:
- Python 3.x
- Pygame library

5
3. System Requirements
3.1 Functional requirements

Functional Requirements:
- The system shall allow the player to move left and right.
- The system shall allow the player to shoot bullets.
- The system shall spawn enemies at timed intervals.
- The system shall detect collisions between bullets and enemies.
- The system shall track and display the player's score.
- The system shall end the game if an enemy collides with the player.
4.External Interface Requirements
4.1 User Interfaces
The user interacts through a keyboard interface:
- LEFT arrow: Move jet left
- RIGHT arrow: Move jet right
- SPACE: Shoot bullets

Game screens:
- Intro Screen: Title, instructions
- Game Screen: Jet, enemies, bullets, score
- Game Over Screen: Final score and retry option
4.2 Hardware Interfaces
CyberShooter requires:
- Keyboard for input
- Monitor for display
- CPU and RAM for processing
4.3 Communications Interfaces
No external communication interfaces are required.
4.4 Software Interfaces
Dependencies:
- Python 3.x for program execution
- Pygame for graphics and event handling

6

5. Non-Functional Requirements
5.1 Performance Requirements
The game shall run at 60 FPS on a standard PC. Enemy spawning and collision detection must
respond within 50ms to ensure smooth gameplay.
5.2 Safety Requirements
The software shall not harm the user's system. It does not access files or the internet.
5.3 Security Requirements

The game shall not store personal information. Security risks are minimal since no networking is
used.

5.4 Software Quality Attributes

Qualities include:
- Usability: Simple controls.
- Maintainability: Code structured in functions.
- Reliability: Stable gameplay without crashes.
- Portability: Runs on any OS with Python & Pygame.

5.5 Other Requirements

Future improvements may include:
- Power-ups and advanced levels
- Background music and sound effects
- Multiplayer support
- Enhanced graphics

7

6. Design / Flow chart

Start Game
Show Menu
Player Input
Update Game
Collision?
Update score
Win / Lose?
End Game
Exit
Play
(Keyboard Controls)
(Player moves, enemies
move, bullets fire)
No Back to update
No Back to update
Yes
Yes

References